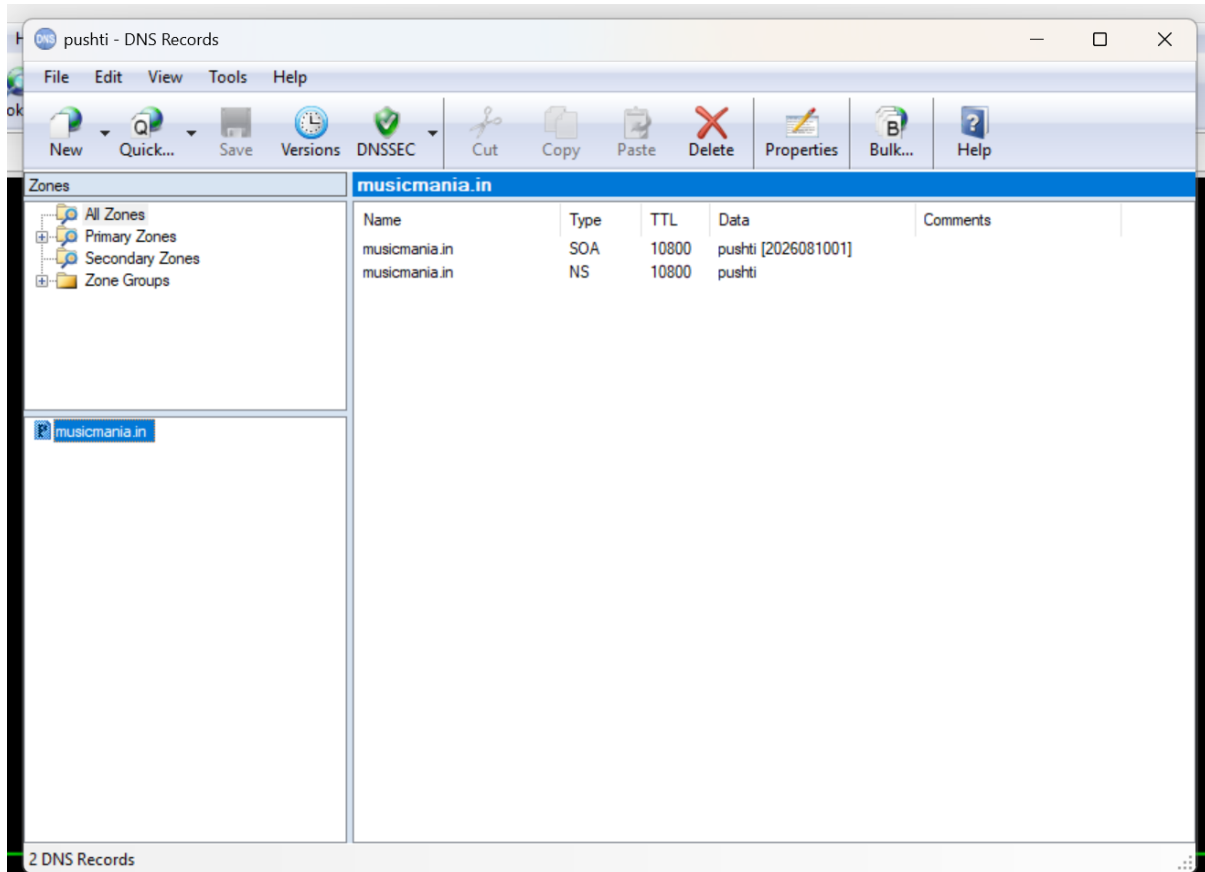


1.Install and set up a DNS server on your local machine or a VM using BIND (Linux) or Simple DNS Plus (Windows), then create a new DNS zone for the domain musicmania.in



2.Add an A record in your DNS server configuration to map www.musicmania.in to the IP address 192.168.1.50, then use the nslookup or dig command to verify the mapping works.

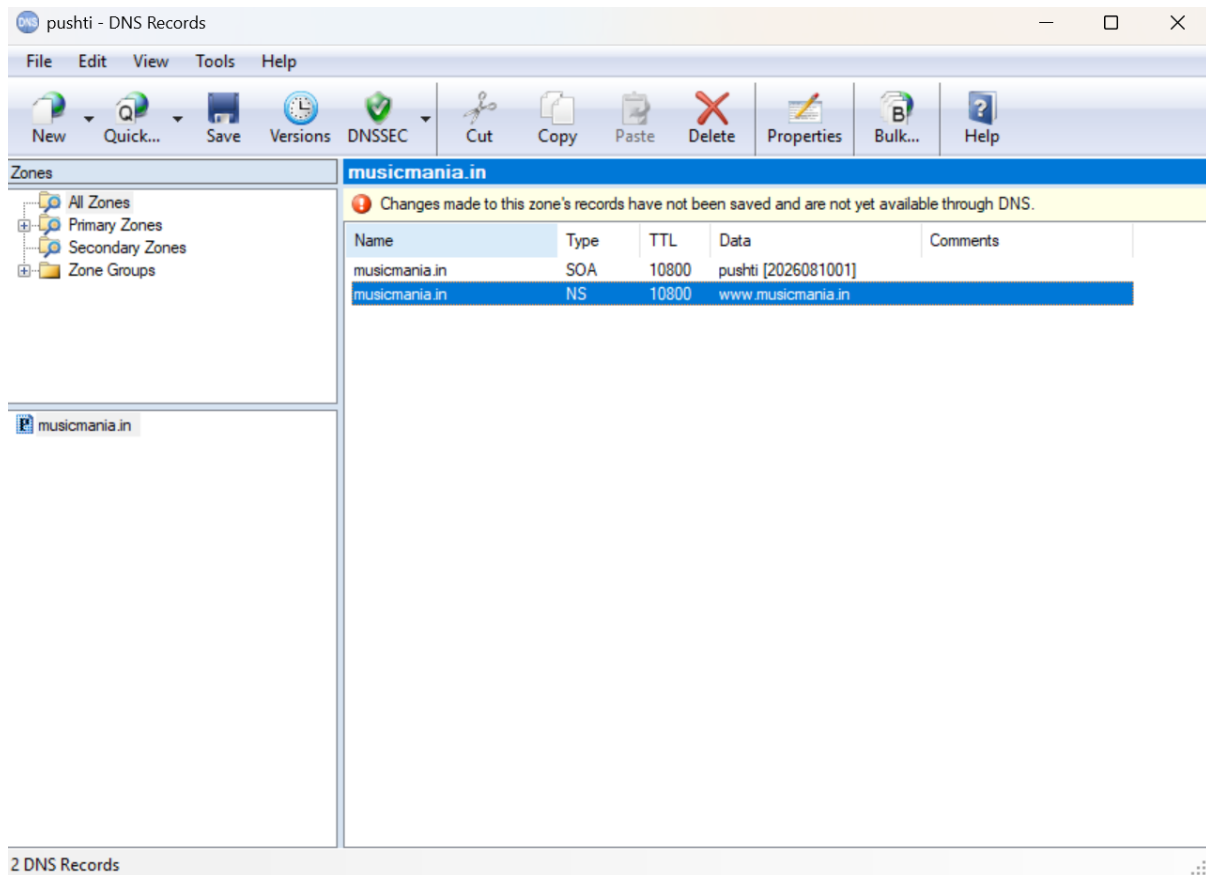
```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\hell0> nslookup www.musicmania.in
Server: UnKnown
Address: fe80::26de:8aff:fe79:8d11

** UnKnown can't find www.musicmania.in: Non-existent domain
```

3. Simulate a typo by adding a CNAME record for play.musicmania.in that points to www.musicmania.in, then test if accessing play.musicmania.in resolves to the correct IP address.



4. Your DNS server is not resolving the domain musicmania.in as expected. Document the step-by-step troubleshooting process you would follow, including checking configuration files, restarting the DNS service, and testing with nslookup or dig.
Hint: Include at least three different troubleshooting commands or checks in your documentation.

If the DNS server is not resolving musicmania.in correctly, first check the **DNS configuration files and zone file** for errors. Then check whether the BIND DNS service is running properly.

Next, restart the DNS service after making any changes and test the domain using nslookup or dig. DNS logs can also be checked to find errors.

Troubleshooting commands/checks:

- `named-checkconf` – checks DNS configuration.
- `named-checkzone` – checks the DNS zone file.
- `systemctl status bind9` – checks whether DNS service is running.
- `systemctl restart bind9` – restarts the DNS service.
- `nslookup` / `dig` – tests DNS name resolution.